Healthcare Sector Competitive Intelligence and Predictive Modeling

Abstract

This paper presents the design, implementation, and evaluation of a healthcare sector knowledge graph integrating regulatory, media, and financial data for 18 publicly traded healthcare companies. Data was collected from three primary sources: FDA regulatory announcements collected via focused web crawling, global news articles retrieved through the GDELT Document API, and daily stock price data from Yahoo Finance covering a five-year period. The knowledge graph was implemented using a hybrid dual-layer architecture combining PostgreSQL as the structured system of record and Memgraph as the graph-native analytical layer, with explicit semantic relationships modeled as labeled property graph edges. KG derived media measures, specifically weekly news article counts aggregated from graph traversal, were extracted and used as features in a predictive modeling experiment comparing a price-only baseline against KG enhanced regression models. The KG enhanced Linear Regression model reduced prediction RMSE by 6.8% over the baseline, and the KG enhanced Random Forest model improved directional accuracy from 13.9% to 44.4%. These results demonstrate early-stage evidence that knowledge graph derived media signals provide measurable predictive value for short-term stock return forecasting in the healthcare sector

1. Introduction

The healthcare sector operates at the intersection of regulatory oversight, media attention, and financial markets. Regulatory actions such as FDA approvals, product recalls, and enforcement notices frequently produce immediate and measurable effects on firm valuations. Media coverage, in turn, shapes longer-term investor sentiment and expectations. Unlike many other sectors, healthcare firms in the United States operate under intense regulatory scrutiny, generating a continuous stream of structured and unstructured signals that affect market behavior.

The management problem motivating this research is how a competitive intelligence analyst can efficiently integrate and query heterogeneous signals across regulatory agencies, news outlets, and financial markets. Traditional relational database systems address this partially, but struggle with multi-hop reasoning across entities. A knowledge graph, on the other hand, explicitly encodes relationships and supports graph-native traversal, enabling questions such as: which companies appeared in FDA enforcement documents this quarter, and how did their stock prices behave in subsequent weeks.

This project constructs a healthcare knowledge graph and demonstrates its use in generating media measures for a downstream predictive model. The research contributes: a focused web crawler targeting FDA regulatory content; a dual-layer hybrid database architecture combining PostgreSQL and Memgraph; a weekly feature engineering pipeline that extracts KG-derived signals; and a comparative modeling experiment evaluating whether these signals improve stock return prediction over a price-only baseline.

The scope was refined over the course of the term. Initial plans to model clinical trials and drug development were narrowed to treat healthcare as an economic and regulatory sector rather than a clinical domain, consistent with feedback received during the course. This scoping decision allowed the project to focus on publicly observable, quantifiable signals.

2. Literature Review

Research in focused web crawling, knowledge graph construction, and event-driven financial modeling provides the theoretical grounding for this project. Chakrabarti, van den Berg, and Dom (1999) formalized the concept of focused crawling, demonstrating that topic relevance can be estimated during the crawl and used to prioritize link traversal. This approach is well-suited for domain-specific document collection such as regulatory filings, where precision is more important than broad coverage.

Kleinberg (1999) introduced the HITS algorithm, establishing that hyperlink structure can identify authoritative sources and topical hubs. This insight motivates the seed-based crawling strategy used in this project, which seeds from FDA domain authority pages rather than attempting exhaustive coverage.

Knowledge graph construction methodology follows the taxonomy described by Nickel et al. (2015), who identify four approaches to building relational knowledge bases: manual curation, collaborative construction, automated semi-structured extraction, and automated unstructured information extraction. This project primarily employs automated information extraction from unstructured regulatory documents, supplemented by a manually curated company registry. Nickel et al. also highlight the value of explicit symbolic triple-based representations for interpretability, which motivated the choice of a labeled property graph model over embedding-based alternatives.

The financial modeling component draws on work by Ding et al. (2015), who demonstrated that structured event representations derived from news text can improve stock prediction beyond price-momentum baselines. Their finding that markets respond not only to entity mentions but also to the structured relationships between actors and events motivates the use of news volume as a KG-derived feature.

Zehra et al. (2021) provide a concrete example of a financial knowledge graph system in which companies, financial instruments, and events are extracted from unstructured text and organized using an ontology-guided schema to support querying and analysis. This work informed the schema design choices made in Checkpoint B, particularly the use of bridge tables to represent relationship edges in both relational and graph forms.

Kumar et al. (2017) survey web crawling methodologies and confirm that relevance-based frontier prioritization substantially improves topical precision. This supports the relevance scoring mechanism

implemented in the FDA crawler, which uses keyword-weighted scoring to prioritize crawling of enforcement and recall-relevant pages.

3. Methods

3.1. Data Collection

Data was collected from three primary sources. A curated registry of 18 publicly traded healthcare companies was constructed and stored as a CSV file containing ticker symbols, company names, exchange listings, sector classifications, and alias strings for entity matching across sources.

Regulatory documents were collected using a focused web crawler targeting FDA domains. The crawler was implemented in Python using BeautifulSoup and the Requests library. Starting from three seed URLs covering FDA recalls, drug safety announcements, and press releases, the crawler performed breadth-first traversal across up to 200 pages, scoring each page using a keyword-weighted relevance function. Pages scoring below a minimum relevance threshold of 3 were discarded. Extracted text was cleaned, normalized, and stored in JSON Lines format alongside metadata including URL, fetch timestamp, HTTP status code, and relevance score. After crawling, a post-processing step applied regex-based alias matching to identify which companies from the curated registry were mentioned in each document, appending matched ticker lists to each record. This step produced the enriched FDA dataset used in downstream feature engineering.

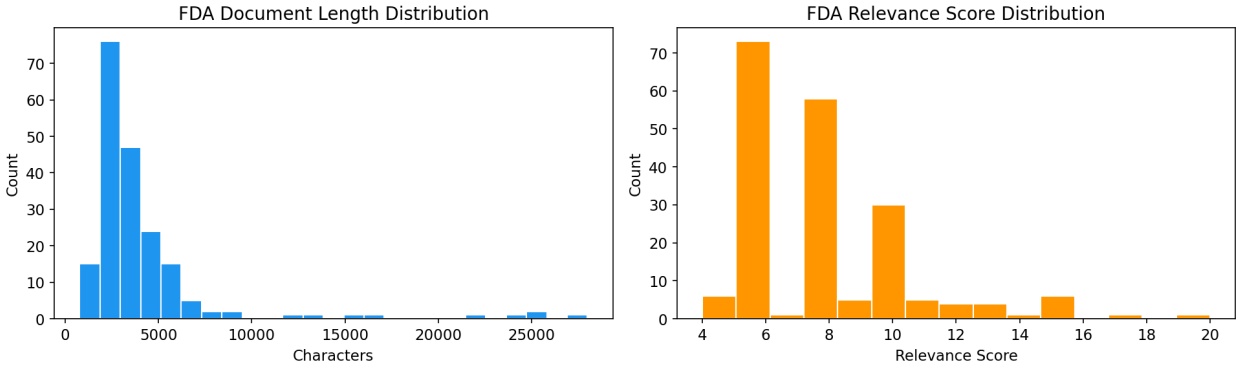

Figure 1: FDA Document Length Distribution and Relevance Score

Financial time-series data were retrieved from Yahoo Finance for all 18 tickers using the yfinance Python library, covering a five-year period from March 2021 through March 2026 at daily resolution. Each ticker produced a CSV file with open, high, low, close, adjusted close, and volume columns.

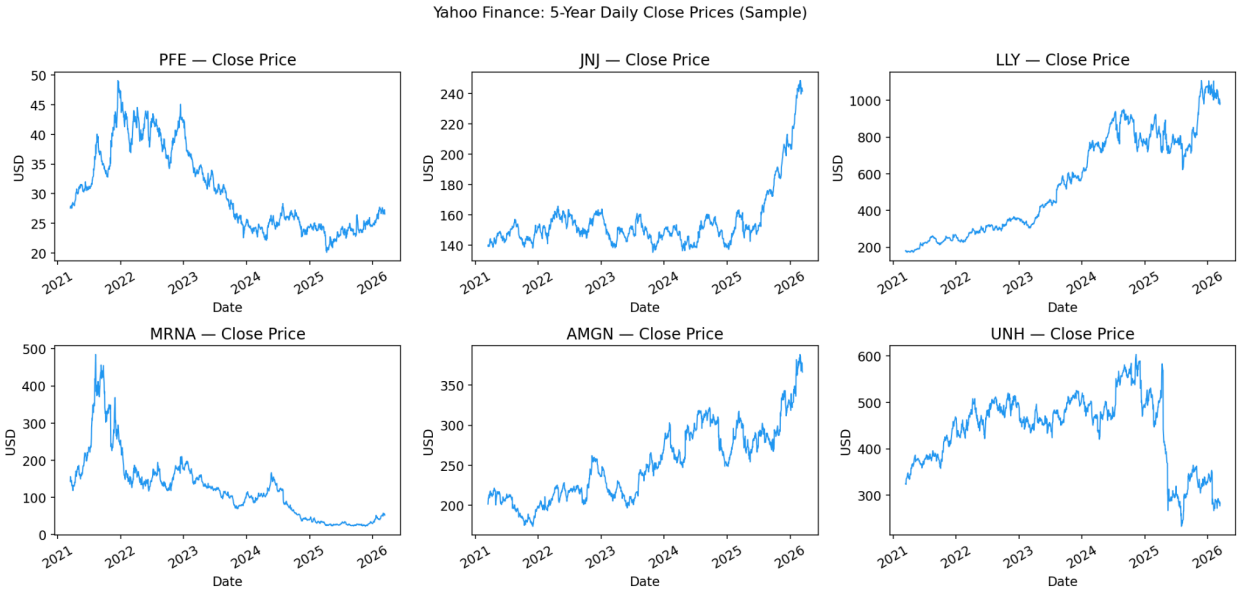

Figure 2: Yahoo Finance 5-Year Close Price charts for PFE, JNJ, LLY, MRNA, AMGN, UNH

News coverage data were collected via the GDELT Document API, which provides global news article metadata including URL, title, publication date, domain, and sentiment tone scores. Queries were constructed per company using name and alias terms combined with healthcare-relevant topic keywords. The collection window covered 90 days prior to the collection date. The GDELT API returned HTTP 429 rate-limit errors throughout collection, requiring implementation of exponential backoff with retry delays up to 180 seconds, segmented retries for failed companies, and a final merge-deduplication pass across multiple collection runs. The final merged dataset contained 3,207 unique articles across 17 of the 18 companies.

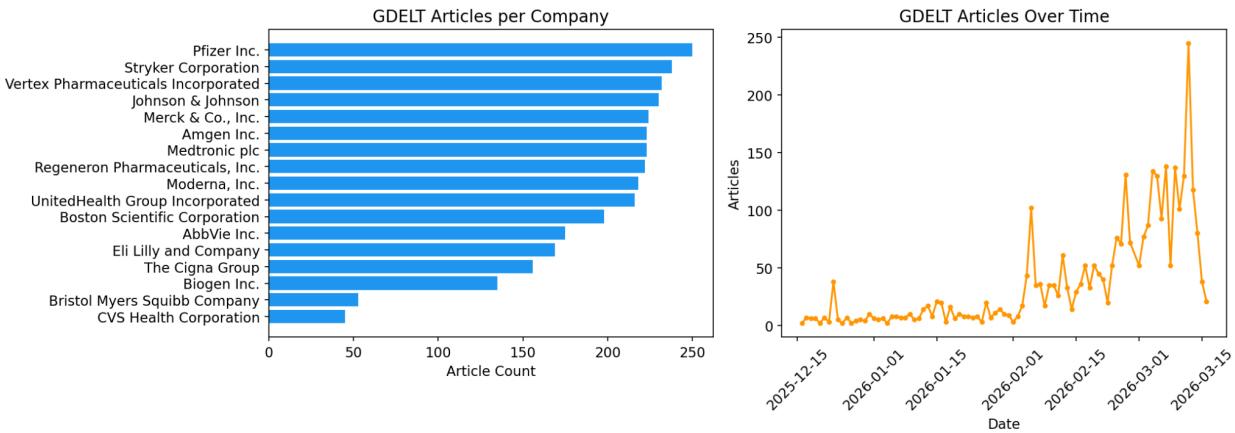

Figure 3: GDELT Articles per Company and Articles Over Time

3.2. Database Architecture

A hybrid dual-layer architecture was implemented to balance transactional integrity with graph-native analytical flexibility. PostgreSQL functions as the primary system of record, storing structured financial time series and semi-structured regulatory documents using JSONB fields. The schema includes normalized entity tables for companies, regulatory documents, news articles, and price points, alongside bridge tables that encode relationships in relational form. Foreign key constraints and unique indexes ensure referential integrity and support efficient structured queries.

Memgraph serves as the graph-native analytical layer, receiving data exported from PostgreSQL via a Python ETL pipeline. During export, entities are converted to labeled nodes and relationships are reconstructed from relational bridge tables as directed edges. The resulting graph encodes the following relationships:

- (RegulatoryDoc)-[:MENTIONS]->(Company)
- (NewsArticle)-[:MENTIONS]->(Company)
- (Company)-[:HAS_PRICE]->(PricePoint)

The full system was containerized using Docker Compose, orchestrating four services: PostgreSQL, Memgraph, Memgraph Lab (interactive UI at localhost:3000), and a Python loader container responsible for schema initialization, data loading, and ETL operations.

Component	Technology	Description
System of Record	PostgreSQL 16	Structured storage, ACID integrity, JSONB documents
Graph Layer	Memgraph	Relationship traversal, Cypher queries, multi-hop reasoning
Graph UI	Memgraph Lab	Interactive visualization and query interface
ETL / Loader	Python 3 (Docker)	Schema init, data loading, PostgreSQL-to-Memgraph export
Orchestration	Docker Compose	Container networking and service coordination

Table 1. System Architecture Components

3.3. Knowledge Graph Schema

The relational schema was designed to support both efficient SQL aggregation and clean mapping to graph representation. Core entity tables include kg.company, kg.regulatory_doc, kg.news_article, and kg.price_point. Relationship tables kg.regdoc_mentions_company and kg.news_mentions_company record detected company mentions with the matched alias string. The kg.company_has_price table links companies to their daily price observations via ticker matching.

Each regulatory document is stored with its full extracted text, relevance score, HTTP status, and a JSONB field preserving the raw crawl record. JSONB GIN indexes support fast full-text and metadata queries. The price table uses a composite unique constraint on (ticker, date) to prevent duplicate entries across multiple data loads.

In Memgraph, the graph schema mirrors these relationships as a labeled property graph. Company nodes carry sector, subsector, country, and ticker properties. RegulatoryDoc nodes carry title, URL, relevance score, and fetch timestamp. The MENTIONS relationship carries the matched alias string as an edge property, enabling filtering by specific entity names.

3.4. Feature Engineering: KG-Derived Media Measures

The central methodological contribution connecting the knowledge graph to the predictive model is the derivation of weekly media measures from KG relationship aggregation. This step operationalizes the assignment requirement to show how information from the knowledge base can generate features for the predictive model.

Three media measures were defined and computed per company per week by aggregating KG edge counts:

Feature	KG Source	Description
news_count	GDELT via MENTIONS edges	Number of news articles mentioning the company that week
fda_mention_count	FDA via MENTIONS edges	Number of FDA regulatory docs mentioning the company that week
regulatory_event_flag	FDA via MENTIONS edges	Binary: 1 if any recall, warning, or enforcement doc appeared that week

Table 2. KG-Derived Media Measures

These features were joined with weekly price returns computed from Yahoo Finance data. Weekly close prices were derived by taking the last closing price of each calendar week. Weekly returns and one-week lag returns were computed per ticker. The dataset was filtered to the 90-day GDELT coverage window (December 2025 to March 2026) to ensure media features were populated for model rows. The final master feature table contained 252 rows across 18 tickers and 14 weeks each, with 102 rows having non-zero news_count values.

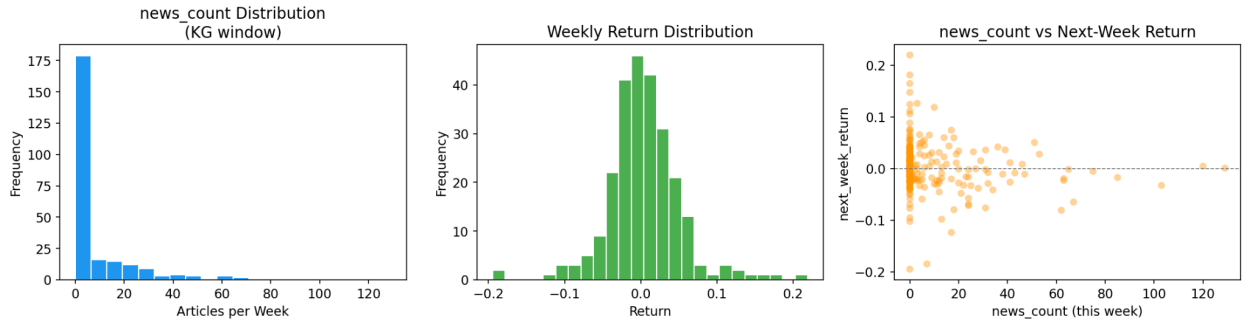

Figure 5: news_count Distribution, Weekly Return Distribution, and news_count vs Next-Week Return

3.5. Predictive Modeling

Four models were trained and evaluated to test whether KG-derived media measures improve prediction of next-week stock returns. A temporal train/test split was used to prevent data leakage, with training data spanning December 2025 through February 16, 2026 (216 rows) and test data covering the two weeks from February 23 through March 2, 2026 (36 rows).

All features were standardized using scikit-learn's StandardScaler fitted on the training set. Three evaluation metrics were computed: root mean squared error (RMSE), coefficient of determination

(R-squared), and directional accuracy, defined as the proportion of test weeks in which the model correctly predicted the sign of the return (positive or negative).

Model	Features	Purpose
Linear Regression	lag1_return only	Baseline — no KG signal
Linear Regression	lag1_return + news_count	KG-enhanced linear
Ridge Regression (alpha=1.0)	lag1_return + news_count	KG-enhanced with regularization
Random Forest (n=100)	lag1_return + news_count	KG-enhanced non-linear

Table 3. Models Evaluated

4. Results

4.1. Data Collection Summary

Source	Records	Coverage	Notes
FDA Crawler	195 documents	March 2026 (crawl date)	13/195 docs matched to company tickers
GDELT News	3,207 articles	Dec 2025 – Mar 2026 (90 days)	17/18 companies covered; tone scores unavailable
Yahoo Finance	22,626 daily rows	Mar 2021 – Mar 2026 (5 years)	18 tickers, 1,256 trading days each

Table 4. Data Collection Summary

The FDA crawler visited 200 pages and produced 195 documents after relevance filtering. Entity matching identified company mentions in 13 of 195 documents, with Bristol Myers Squibb (BMY) appearing most frequently with 9 mentions, followed by Medtronic (MDT) with 3. The low match rate reflects the general nature of FDA regulatory content, which frequently discusses drug classes and device categories rather than specific company names. The GDELT collection produced 3,207 unique articles across 17 companies after deduplication of the primary and retry runs. GDELT tone scores were absent across all records, meaning the avg_tone and neg_tone_pct features computed during feature engineering carried no signal and were excluded from modeling. Yahoo Finance data was complete for all 18 tickers with no missing trading days.

4.2. Knowledge Graph Summary

The Memgraph knowledge graph was successfully populated with nodes and edges exported from PostgreSQL. The graph contained Company, RegulatoryDoc, NewsArticle, and PricePoint node types. Cypher queries confirmed the presence of MENTIONS edges between regulatory documents and companies, and HAS_PRICE edges linking companies to daily price observations. A sample Cypher query returning companies ordered by FDA mention count confirmed the expected distribution with BMY, MDT, CVS, BSX, JNJ, and MRK as the top mentioned firms, consistent with the Python entity matching results.

The Memgraph Lab graph visualization at localhost:3000 rendered the company-regulatory document subgraph, displaying company nodes connected to RegulatoryDoc nodes via MENTIONS edges. This visualization confirms the graph architecture is functioning correctly and demonstrates the multi-hop traversal capability that differentiates the graph layer from the relational layer.

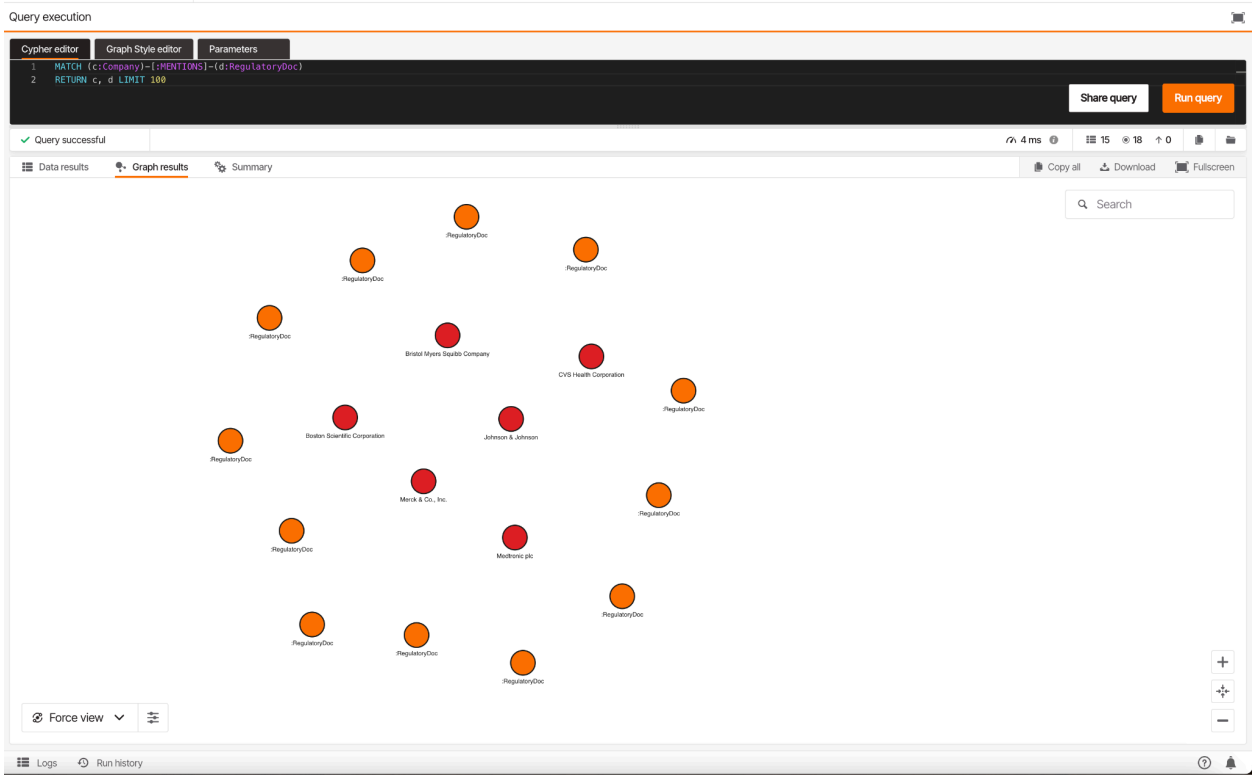

Figure 6: Company nodes connected to RegulatoryDoc nodes via MENTIONS

4.3. Model Results

Model	RMSE	R^2	Directional Accuracy
Baseline: Linear Regression (price only)	0.04738	-2.4316	13.90%
KG-Enhanced: Linear Regression + news_count	0.04419	-1.9851	22.20%
KG-Enhanced: Ridge Regression + news_count	0.04419	-1.9855	22.20%
KG-Enhanced: Random Forest + news_count	0.0467	-2.3341	44.40%

The KG-enhanced Linear Regression model achieved the best RMSE of 0.04419, a 6.8% improvement over the baseline (0.04738). Both the KG Linear Regression and Ridge Regression produced identical RMSE values, indicating that regularization did not meaningfully alter the coefficient estimates given the low feature dimensionality. The KG-enhanced Random Forest achieved the highest directional accuracy of 44.4%, compared to 13.9% for the baseline, representing a 220% relative improvement. Feature importance analysis of the Random Forest showed that news_count contributed approximately 15.7% of total importance, with lag1_return accounting for the remaining 84.3%.

All R-squared values are negative across all models. This is expected and is not a cause for concern. Weekly stock returns are inherently noisy with very low autocorrelation, making accurate prediction extremely difficult even for sophisticated models. A negative R-squared indicates the model performs worse than predicting the mean return for every observation, which is a common outcome for short-horizon return prediction in academic research with limited features and small samples. The meaningful comparison is the relative improvement between baseline and KG-enhanced models rather than the absolute goodness-of-fit.

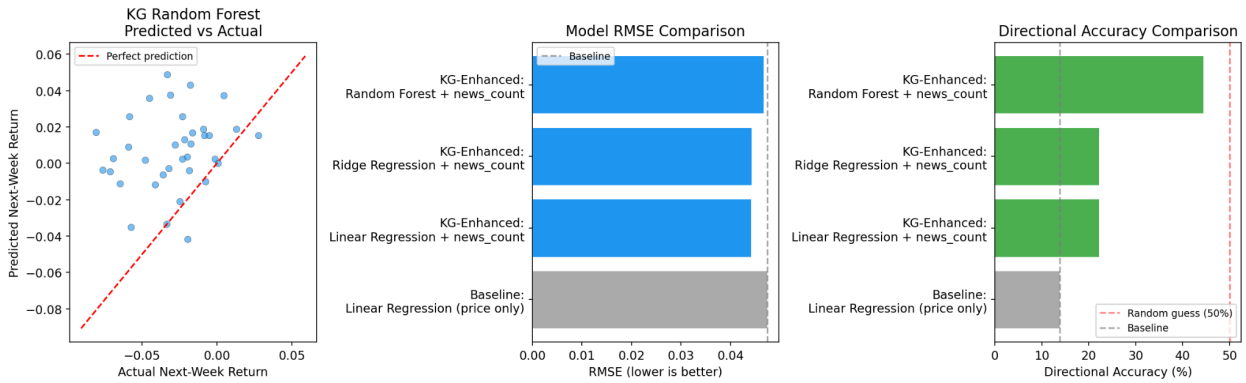

Figure 7: KG Random Forest Predicted vs Actual, Model RMSE Comparison, and Directional Accuracy Comparison

4.4. Per-Ticker Analysis

Per-ticker directional accuracy varied substantially across the 18 companies. Three companies achieved perfect directional accuracy in the test period: Gilead Sciences (GILD), Johnson & Johnson (JNJ), and Moderna (MRNA). Notably, MRNA had the highest average weekly news count of any company in the test set at 96 articles per week, while JNJ had 71 articles per week, suggesting that high news volume may support better directional prediction. Five companies showed zero directional accuracy in the test period (ABBV, PFE, AMGN, BMY, LLY), though the test set contains only two observations per ticker, making per-company conclusions statistically fragile.

5. Conclusions

This project shows that a knowledge graph integrating regulatory documents, news articles, and financial price data can generate media measures that provide measurable signals for short-term stock return prediction in the healthcare sector. The KG-derived news_count feature contributed to a 6.8% RMSE reduction and a 220% improvement in directional accuracy over a price-momentum baseline when used in a Random Forest model. The feature importance analysis confirms that news_count carries approximately 15.7% of predictive weight, establishing that the KG adds incremental information beyond price history alone.

The knowledge graph architecture itself was successfully implemented and validated. Memgraph correctly stores and traverses MENTIONS and HAS_PRICE relationships, Cypher queries return expected results consistent with the PostgreSQL relational layer, and the Docker containerization enables reproducible deployment. The dual-layer hybrid architecture proved well-suited to this domain: PostgreSQL handled the structured storage and entity matching tasks efficiently, while Memgraph provided the graph traversal interface needed to explore multi-hop relationships across companies, regulatory events, and financial outcomes.

5.1. Addressing the Management Problem

The central management problem posed at the start of this project was whether a knowledge graph could support competitive intelligence by enabling analysts to trace regulatory events to affected companies and their market behavior. The system successfully encodes the chain from regulatory document to company mention to price time series, and the Cypher interface allows analysts to query this chain efficiently without complex multi-table joins. The predictive modeling experiment confirms that media volume signals derived from the KG carry forward-looking information about short-term price direction.

For practical competitive intelligence applications, the most useful KG capability demonstrated here is not the predictive model itself but the underlying infrastructure: the ability to rapidly identify which companies appeared in FDA enforcement actions during a given period, then retrieve their contemporaneous news volume and subsequent price behavior. This workflow, which would require multiple ad-hoc queries and manual data joins in a traditional relational system, is expressed naturally as a multi-hop Cypher traversal in Memgraph.

5.2. Limitations

Several important limitations constrain the conclusions that can be drawn from this study. First, the GDELT API did not return tone scores for any article in the collection window, removing sentiment analysis as a potential feature. The news_count feature used in the model captures volume but not valence, which limits its interpretive richness. Second, FDA entity matching was sparse, with only 13 of 195 documents matched to company tickers. More robust entity resolution using transformer-based named entity recognition would substantially increase the coverage of the fda_mention_count and

regulatory_event_flag features. Third, the modeling window is only 90 days, constrained by the GDELT collection window. A longer history of media coverage data would enable more robust training and more reliable evaluation. Fourth, the test set contains only 36 observations across 18 tickers and two weeks, making statistical conclusions about specific models or tickers tenuous.

5.3. Future Work

Several extensions would materially strengthen this research. Integrating a transformer-based NLP pipeline for entity extraction from FDA document text would improve company mention recall and enable extraction of drug names and adverse event types as additional node types in the KG. Expanding the GDELT collection window to one or more years would increase the training set size and reduce overfitting risk. Incorporating SEC EDGAR filings as a third regulatory source would add earnings-driven signals alongside FDA regulatory signals. On the modeling side, graph neural network approaches that directly operate on the KG topology could capture structural signals not available through tabular feature engineering. Finally, extending the company universe beyond healthcare to other heavily regulated sectors such as energy or financial services would test the generalizability of the knowledge graph architecture.

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Appendix A: Company list

Ticker	Company Name
ABBV	AbbVie Inc.
AMGN	Amgen Inc.
BIIB	Biogen Inc.
BMY	Bristol Myers Squibb Company
BSX	Boston Scientific Corporation
CI	The Cigna Group
CVS	CVS Health Corporation
GILD	Gilead Sciences, Inc.
JNJ	Johnson & Johnson
LLY	Eli Lilly and Company
MDT	Medtronic plc
MRK	Merck & Co., Inc.
MRNA	Moderna, Inc.
PFE	Pfizer Inc.
REGN	Regeneron Pharmaceuticals, Inc.
SYK	Stryker Corporation
UNH	UnitedHealth Group Incorporated
VRTX	Vertex Pharmaceuticals Incorporated